

Problem Set 7 ^{11, 1, 5, 7, 9}
↑
a and e

11.1

a.) 1.) X_e decreases and X increases. At higher temperatures conversion increases ~~but~~ but X_e decreases because at equilibrium the temperature pushes it towards the reactants, causing a lower X_{eq} .

2.) T_0 has the greatest effect on the temp profile. E has a greater effect on conversion.

3.) As T_0 increases the max rate occurs earlier.

4.) Temperature has a very large effect on conversion and rates. Combining this with other factors such as the Flow rate ~~and~~ can effect the yield of your reactor.

e.) 1.) $220 \frac{\text{kcal}}{\text{s}} \times \frac{\text{mol} \cdot \text{K}}{0.018 \text{ kcal}} \times \frac{1}{(400 - 270) \text{ K}} = 94 \frac{\text{mol}}{\text{s}}$

2.) $Q = n C_p dT = 94 \times 18 (400 - 270) = 220 \text{ kW/s}$

$A = \cancel{Q_{max}} \frac{220 \times 10^3}{100 (460 - 400 - 350 + 270)} \times \ln \left(\frac{460 - 400}{350 - 270} \right)$
 $= 31.6 \text{ m}^2$

11.5

P11-S

$$F_A W = F_A W_{in} - F_A W_{out} \quad \Delta W = 0$$

$$\frac{dF_A}{dW} = -r_A = 0$$

$$X = \frac{F_A - F_{A0}}{F_{A0}}$$

$$F_{A0} \frac{dX}{dW} = -r_A = 0$$

$$\frac{dX}{dW} = \frac{-r_A}{F_{A0}}$$

$$-r_A = k C_A$$

$$C_A = C_{A0} \frac{(1-X) T}{(1+X) T_0}$$

$$\text{bc } \varepsilon = 1 \times (2-1) = 1$$

$$C_B = C_C = \frac{C_{A0} X T}{(1+X) T_0}$$

$$k = .133 \exp \left[\frac{31400}{8.314} \left(\frac{1}{450} - \frac{1}{T} \right) \right]$$

$$\text{Max } W = 50 \text{ kg} \quad \Delta P = 0$$

$$C_{A0} = \frac{P_0}{RT_0} = .271 \frac{\text{mol}}{\text{dm}^3}$$

40

$$\text{Energy balance: } X(20 \text{ kJ/mol}) = [(1-X)C_{PA} + X(C_{PB} + X C_{PC})] (T - T_0)$$

$$X 20000 \text{ J/mol} = 40 (T - 450)$$

$$T = 450 + 500 X$$

At around 850 K a conversion of 80% was reached. T

varies linearly w/ X from the energy balance so T vs X is a straight line

2.) As T decreases X decreases if catalyst wt stays the same

$$3.) \frac{dQ}{dV} = \underbrace{\sum F_i C_{Pi} \frac{dT}{dW}}_0 + (-r_A) (-\Delta H_{rxn})$$

$$k = .133 \quad -r_A = .133 \times .271 \times \frac{1-X}{1+X}$$

$$\frac{dX}{dW} = \frac{-r_A}{F_{A0}}$$

From the plot, as V inc Q removed decreases.

$$4 \text{ and } 5.) \quad P = \frac{P_0}{P}$$

$$\frac{dP}{dW} = \frac{\alpha(1+X)T}{2P T_0}$$

multiply concentrations by P as well

$$\alpha = .019$$

T vs W plot increases but at a slower rate as time goes on. X vs W is the same, with a max

conversion of ~ 0.17 . P vs W is a straight line, reaching $\sim P 1.5$.

6.) Ways that this problem could be worked incorrectly is by not accounting for epsilon in the concentrations.

P 11-7

$A \leftrightarrow B$ PBR w/ $P \neq 1$

$$y_{A0} = \frac{2}{3} \quad y_{I0} = \frac{1}{3} \quad F_{A0} = \frac{5 \text{ mol}}{\text{min}} \quad T_0 = 300 \quad C_{A0} = 2 \frac{\text{mol}}{\text{dm}^3}$$

$$V = \frac{W}{V_B} \quad r_A = P_B r_A \quad \text{no } \epsilon$$

$$\frac{dF_A}{dW} = -r_A$$

$$F_{A0} \frac{dX}{dW} = -r_A' \quad W = V P_B \quad r_A = P_B r_A'$$

$$F_{A0} P_B \frac{dX}{dV} = \frac{r_A'}{P_B}$$

$$\frac{dX}{dV} = \frac{-r_A'}{F_{A0} P_B}$$

$$\frac{dP}{dV} = \frac{2 P P_B T}{P T_0}$$

$$-r_A = k \left(C_A - \frac{C_B}{K_C} \right)$$

$$C_A = C_{A0}(1-X) P \frac{T}{T_0} \quad C_B = C_{A0} X P \frac{T}{T_0}$$

$$-r_A = k C_{A0} P \frac{T}{T_0} \left(1-X - \frac{X}{K_C} \right)$$

From textbook and notes $X_{eq} = \frac{K_C}{1+K_C}$

Energy balance: Adiabatic and no work

$$0 = \sum F_{A0} h_A(T_0) - \sum F_A h_A(T)$$

$$T = \frac{[T_0 \Delta_r C_p - \Delta_r H(T_0)] X + T_0 \sum \theta_i C_{Pi}}{\Delta_r C_p X + \sum \theta_i C_{Pi}}$$

$$\Delta_r C_p = 20(160) = -140 \frac{\text{cal}}{\text{mol K}}$$

$$\sum \theta_i C_{Pi} = \frac{1}{2} \times 18 + 160 = 169 \frac{\text{cal}}{\text{mol K}}$$

T vs V is an almost linear curve, reaching about 335 @ 40 dm³

as it is rising w/ X .

X vs V is rising w/ $X(40) = \sim 0.25$. It doesn't reach

X_e . P increases linearly w/ V .

8.) The graphs stay mostly the same.

c.) From energy balance $Q = -r_A \times \Delta H_r \times V$

Plot is increasing as V increases.

P11-9

$$\begin{aligned} K_e(T) &= K_e(T_0) \exp\left(-\frac{\Delta_r H^\circ}{R} \left(\frac{1}{T} - \frac{1}{T_0}\right)\right) \\ &= 50000 \exp\left(\frac{-30000}{1.987} \left(\frac{1}{360} - \frac{1}{323.15}\right)\right) \\ &= 10035 \end{aligned}$$

$$K_e = \frac{x_{eq}}{a - x_{eq}} \Rightarrow x_{eq} = \frac{\sqrt{K_e}}{1 + \sqrt{K_e}} = 0.056$$

$$X = \frac{\sum \theta_i C_{pi} (T - T_0)}{-\Delta^\circ H_{rxn}} = \frac{T - T_0}{600}$$

$$T = T_0 + 600X$$